RAHUL MITRA

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SUMMARY

Data and AI engineer building optimisation, LLM, and data systems for manufacturing, with experience at Abbott, an STMicroelectronics warehouse AI capstone, Amazon Robotics computer vision research, and Churp, a civic platform built with People's Association and PA ICT (GovTech). Final-year NUS Industrial and Systems Engineering and Computer Science student.

EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Software Engineering, Data Structures and Algorithms, Artificial Intelligence, 3D Computer Vision, Computer Graphics, Operations Research, Simulation, Statistics, Linear Algebra.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine reading each site's shelf coordinates, dimensions, and operating rules, so one model fits any STMicroelectronics warehouse without per-site rework.
- Defined a historical-replay evaluation comparing 3 metrics against current allocation (travel distance per order, handling time, and slow-mover Golden Region occupancy), piloting on 2 AMK warehouses: dice and finished goods.
- Formulated a 4-stage shelf-ranking pipeline: movement profiling from pick frequency, dwell time, and demand; filtering on packaging zone, fit, capacity, and restricted areas; Golden Region priority for fast movers; and logged rationale per decision.

Global Operational AI Systems & Data Engineer (Part-Time), Abbott Laboratories (Contract) **Jun 2026 – Present**

- Developed ADAM, an offline tool-calling LLM assistant (Gemma 4 E4B on llama.cpp) for 6 spray dryers across 4 global sites, giving engineers self-serve answers to decisions previously requiring process-control specialists.
- Rebuilt 4 sites' Excel-based Advanced Process Control (APC) models as 7 steady-state gain models solved by HiGHS LPs and SLSQP QPs, returning setpoints, binding constraints, and shadow prices in milliseconds.
- Verified the APC replica against live plant setpoints, matching move direction in 83% of cases and temperatures within 0.12 °C over 24 readings.
- Implemented a Seeq signal-analytics toolkit (Kalman smoothing, spectral and lag analysis, change-point detection, matrix profiles) and a 5-stage offline handbook RAG pipeline (OCR, ONNX embeddings, Chroma, LangChain).
- Enforced Seeq plant-state and 30-minute steadiness checks before each optimiser solve, and kept LLM narration free of computed numbers through schema-constrained tool calls verified against solver results.
- Replaced a reference simulator's bisection search with minimum lexicographic constraint relaxation via LPs, fixing 2 QP defects.
- Proved production modules identical to research notebooks via exact-equality parity tests, plus 20 engine tests and GPU-free CI.

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Engineered a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, comparing 3 approaches (heuristic, MIP, genetic algorithm) to produce decision-ready schedules.
- Validated a 15-stage changeover-data pipeline on 5 years of unseen raw data, with zero failures across operator and machine anomalies.
- Deployed a team-built APC simulator to live internal users on Azure App Service, with Docker, access restrictions, and session hardening.
- Trained regional engineering teams across Asia and Europe on repeatable AI workflows spanning 7 areas: prompting, model selection, benchmarks, cost awareness, agents, validation, and safeguards.
- Strengthened Excel, VBA, Python, master-data, and scheduling workflows with 3 safeguards: validation rules, correction tables, and reproducible intermediate stages, leaving every transformation auditable.
- Documented deployment, access provisioning, and technical handover across 3 procedures, so internal manufacturing and engineer-training users stayed supported after transfer to owning teams.

Software Engineer (Full Stack & Cloud), People's Association (Pro bono) **Sep 2025 – Sep 2026**

- Designed, built, and deployed Churp, a resident-driven garden-plot balloting platform now live in Tengah and slated for national rollout, seeded by a S$20K Sparks Community Innovation Fund award.
- Integrated Singpass Login and Myinfo (OIDC) to verify residents' identity and address, keeping balloting fair for 150 sign-ups at release.
- Provisioned 10 AWS services with Terraform (ECS Fargate, Multi-AZ RDS with PostGIS, WAFv2, Lambda, KMS), deployed through keyless GitHub Actions CI/CD across 5 repos with gated migrations and auto-rollback.
- Owned most software development and deployment across admin and public Next.js frontends, a NestJS backend, and GIS mapping.
- Surveyed garden sites into geospatial datasets for mapping and plot allocation, and designed rules-based balloting across 4 control points (eligibility, conflict resolution, exceptions, admin oversight), keeping allocation auditable.
- Led deployment planning, multi-repo coordination, and stakeholder management in an 11-person team across People's Association, Tengah community partners, and PA ICT (GovTech).

RESEARCH EXPERIENCE

Research Intern, Digital Supply Chain Group (Incoming), A*STAR **Jan 2027 – May 2027**

- Selected for a 4-month research attachment in ARTC's Digital Supply Chain Group, running alongside final-semester studies.

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Tackled an Amazon Robotics brief spanning 3 workstreams (camera ISP, super-resolution, image restoration) to make low-cost cameras rival far pricier sensors, mentored by Zelig CTO Dan Shao (ex-iRobot).

- Reproduced a published UNet super-resolution model in PyTorch from broken author code, recovering its undocumented evaluation protocol and training 12 models, matching reported results at 8x and landing within 0.78 dB at 2x and 4x.
- Diagnosed a robustness gap where fixed-blur super-resolution models lost most PSNR gains under unseen blur, then added randomised-blur training and fine-tuning, recovering up to 0.7 dB across lens, out-of-focus, and resampling blur.
- Launched 3 trained models as a public web app via a Flask inference API on a Colab T4 GPU behind a secure tunnel on a reserved domain.
- Authored and presented a super-resolution literature review spanning multi-image, video, and single-image methods, organised around camera information loss and recovery, comparing 4 model families on fidelity and hallucination risk.

PROJECTS

- Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants, Sep 2026): Directed a 5-person team building decentralised drone-swarm coordination in a deterministic Unreal Engine 5.8 C++ simulation; under a ground-link jam, 30 drones confirmed all 8 targets in 10 of 10 runs, versus an average 2.1 of 8 for an operator-run fleet.
- Personal Project, Hailo Edge AI Training and Deployment (Mar 2026): Converted YOLO26 through transfer learning and the Hailo Dataflow Compiler for document detection on a Hailo-8L rated 13 TOPS at 1.5 W to speed up large-batch scanning.
- Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams, Jan 2026): Built an autonomous loop managing electricity and utility savings across multiple HDB households at once on SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.
- Hackathon Project, Waaah Comics: 24-Hour Gesture-to-Comic Build (Jan 2026): Turned MediaPipe 3D hand landmarks into generated comic strips in a 24-hour sprint with Gemini 2.0 Flash, Veo 3, and Konva.js canvas tooling.
- Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon (Jun 2025): Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.
- Competitor, DSTA BrainHack 2025 CTF (Feb 2025): Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, ranking top 25% of 400+ teams.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Independent Bug Bounty Researcher, YesWeHack **May 2024 – Present**

- Assessed GovTech GBBP12/13, Land Transport Authority, and 13+ other programs for SSRF, CSRF, injection, and access-control flaws, automating recon and disclosure evidence with Burp Suite and Python fuzzing scripts.

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

NATIONAL SERVICE

Republic of Singapore Navy, Naval Chef & Base Support Assistant **Jan 2021 – Jan 2023**

- Compressed a month-long manual analysis of 6 months of food-waste data into a 3-minute Excel VBA report in a restricted environment.

SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, C++, SQL, Bash, Excel VBA
Web and Cloud: React, Next.js, Three.js, NestJS, FastAPI, Flask, Docker, AWS (ECS Fargate, RDS, Lambda, API Gateway, WAFv2), Terraform, AWS SAM, Azure App Service, GitHub Actions and Bitbucket CI/CD, Supabase, Singpass Login and Myinfo (OIDC), API design, GIS mapping
AI and Machine Learning: RAG, AI agents, schema-constrained tool calling, local LLM serving (llama.cpp, Gemma), vector search, LangChain, PyTorch, ONNX, Chroma, BERT/transformer fine-tuning, model evaluation, deep RL (PPO, A2C, DDPG, DQN), GPU training (CUDA) on ASPIRE 2A HPC, Seeq time-series analysis (Kalman filtering, spectral analysis, change-point detection, matrix profile)
Simulation and Operations Research: SimPy, OR-Tools CP-SAT, HiGHS linear programming, SLSQP quadratic programming, constraint relaxation, sensitivity analysis (shadow prices), discrete-event simulation, hybrid flow-shop scheduling
Security: OWASP Top 10, Burp Suite, Wireshark, OAuth2/JWT, SSRF/CSRF/injection testing, fuzzing, secure design
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery